Project Design Phase
Problem – Solution Fit Template

Date	26 June 2026
Team ID	
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	2 Marks

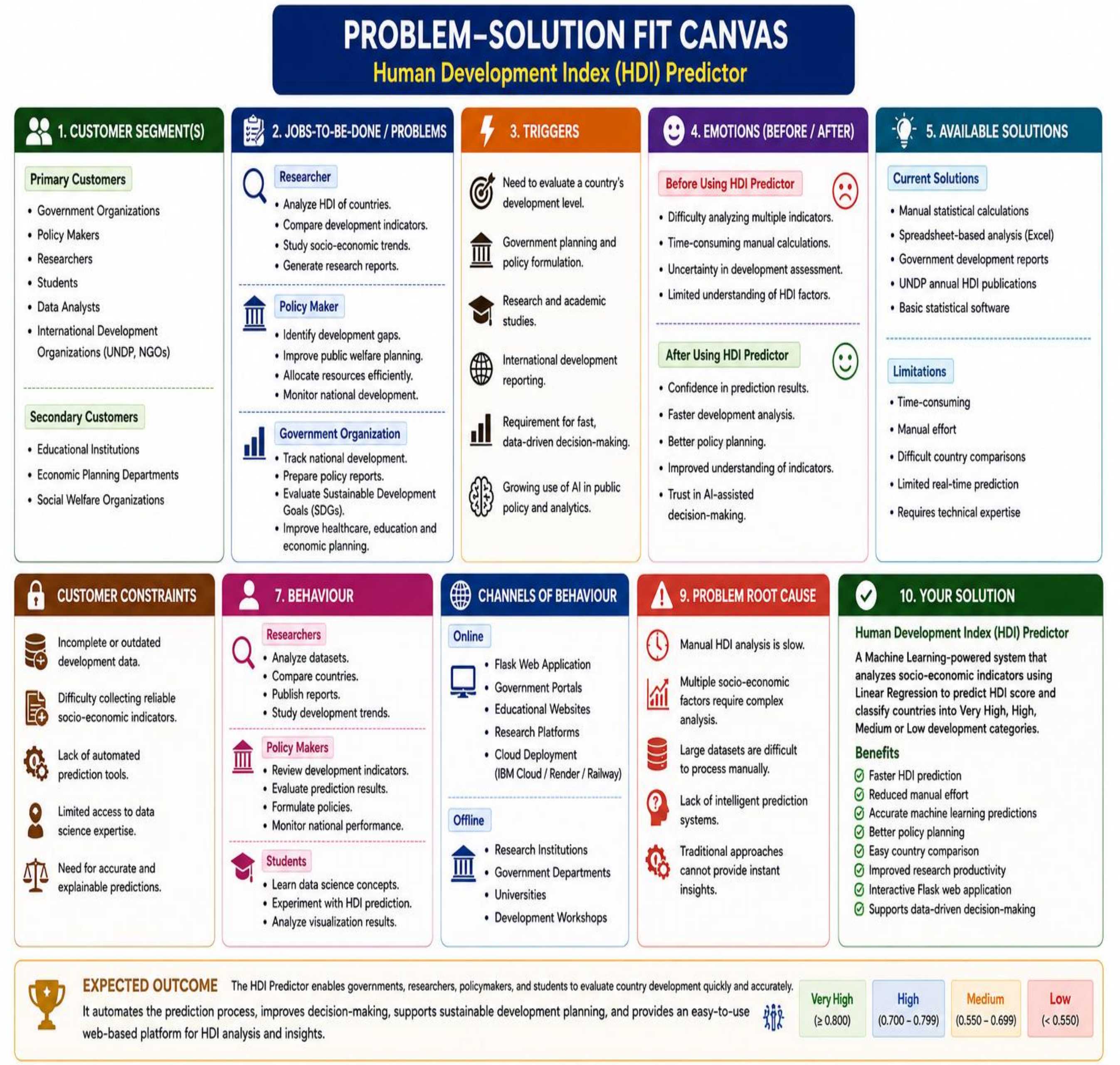
Problem – Solution Fit Template:

The Human Development Index (HDI) Predictor is a Machine Learning-powered web application that predicts the Human Development Index (HDI) score of a country using key socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita. The application performs data preprocessing, predicts the HDI score using a Linear Regression model, classifies the result into Very High, High, Medium, or Low human development categories, and displays the prediction through an interactive Flask web application.

Key Features:

- Automated HDI Score Prediction
- Real-time HDI Category Classification
- Linear Regression Prediction Engine
- Data Preprocessing & Feature Engineering
- Interactive Flask Web Application
- Data Visualization (Heatmaps, Scatter Plots, Distribution Plots)
- Fast and Accurate Development Analysis
- Reduced Manual Analysis and Better Decision-Making

Template:


PROBLEM-SOLUTION FIT CANVAS
Human Development Index (HDI) Predictor
1. CUSTOMER SEGMENT(S)
Primary Customers
• Government Organizations
• Policy Makers
• Researchers
• Students
• Data Analysts
• International Development Organizations (UNDP, NGOs)
Secondary Customers
• Educational Institutions
• Economic Planning Departments
• Social Welfare Organizations
2. JOBS-TO-BE-DONE / PROBLEMS
Researcher
• Analyze HDI of countries.
• Compare development indicators.
• Study socio-economic trends.
• Generate research reports.
Policy Maker
• Identify development gaps.
• Improve public welfare planning.
• Allocate resources efficiently.
• Monitor national development.
Government Organization
• Track national development.
• Prepare policy reports.
• Evaluate Sustainable Development Goals (SDGs).
• Improve healthcare, education and economic planning.
3. TRIGGERS
Need to evaluate a country's development level.
Government planning and policy formulation.
Research and academic studies.
International development reporting.
Requirement for fast, data-driven decision-making.
Growing use of AI in public policy and analytics.
4. EMOTIONS (BEFORE / AFTER)
Before Using HDI Predictor
• Difficulty analyzing multiple indicators.
• Time-consuming manual calculations.
• Uncertainty in development assessment.
• Limited understanding of HDI factors.
After Using HDI Predictor
• Confidence in prediction results.
• Faster development analysis.
• Better policy planning.
• Improved understanding of indicators.
• Trust in AI-assisted decision-making.
5. AVAILABLE SOLUTIONS
Current Solutions
• Manual statistical calculations
• Spreadsheet-based analysis (Excel)
• Government development reports
• UNDP annual HDI publications
• Basic statistical software
Limitations
• Time-consuming
• Manual effort
• Difficult country comparisons
• Limited real-time prediction
• Requires technical expertise
CUSTOMER CONSTRAINTS
Incomplete or outdated development data.
Difficulty collecting reliable socio-economic indicators.
Lack of automated prediction tools.
Limited access to data science expertise.
Need for accurate and explainable predictions.
7. BEHAVIOUR
Researchers
• Analyze datasets.
• Compare countries.
• Publish reports.
• Study development trends.
Policy Makers
• Review development indicators.
• Evaluate prediction results.
• Formulate policies.
• Monitor national performance.
Students
• Learn data science concepts.
• Experiment with HDI prediction.
• Analyze visualization results.
CHANNELS OF BEHAVIOUR
Online
• Flask Web Application
• Government Portals
• Educational Websites
• Research Platforms
• Cloud Deployment (IBM Cloud / Render / Railway)
Offline
• Research Institutions
• Government Departments
• Universities
• Development Workshops
9. PROBLEM ROOT CAUSE
Manual HDI analysis is slow.
Multiple socio-economic factors require complex analysis.
Large datasets are difficult to process manually.
Lack of intelligent prediction systems.
Traditional approaches cannot provide instant insights.
10. YOUR SOLUTION
Human Development Index (HDI) Predictor
A Machine Learning-powered system that analyzes socio-economic indicators using Linear Regression to predict HDI score and classify countries into Very High, High, Medium or Low development categories.
Benefits
Faster HDI prediction
Reduced manual effort
Accurate machine learning predictions
Better policy planning
Easy country comparison
Improved research productivity
Interactive Flask web application
Supports data-driven decision-making
EXPECTED OUTCOME The HDI Predictor enables governments, researchers, policymakers, and students to evaluate country development quickly and accurately.
It automates the prediction process, improves decision-making, supports sustainable development planning, and provides an easy-to-use web-based platform for HDI analysis and insights.
Very High (≥ 0.800)
High (0.700 – 0.799)
Medium (0.550 – 0.699)
Low (< 0.550)